FinAgent : Multi-Agent System for Automated Financial Risk Screening

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1. Background

SEC EDGAR filings include **critical, company-specific yearly data,** combining structured financial metrics and extensive unstructured narrative disclosures

While structured data is widely used for quantitative analysis and **peer benchmarking**, narrative sections are harder to process and compare across companies

2. Motivations & Objectives

Motivation: Lack of **scalable methods** to contextualize **numerical financial signals with narrative risk disclosures** across comparable companies

Aim:

- Merge financial metrics + narrative disclosures for added context
- Identify top-K similar companies
- Compare signals, trends, and risk patterns across peers
- Generate concise, interpretable risk summaries with highlighted red flags

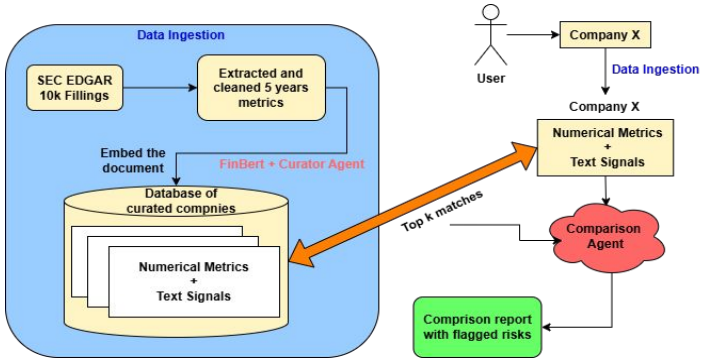
3. Related Work

FinanceBench: Evaluates LLM-based QA on financial documents; highlights challenges in reasoning over long filings and numerical data

FinGPT: Domain-specific LLM for financial text analysis, focusing on summarization, sentiment, and QA over disclosures

Financial NLP: Prior work applies NLP to filings (e.g., sentiment, risk extraction), but limited integration with structured financial metrics for cross-company comparison

4. Methodology



Data Curation:

1. Fetch 5 years of 10K filings per company from SEC EDGAR and pull structured metrics.
2. Parse 6 text sections (Risk, Cybersecurity, Legal, etc) for risk signals.
3. Outputs parsed Company Risk Signals and numeics for next agent.

Finding peers:

1. Curated risk signals and financial delta labels are encoded into dense vectors via a sentence-transformer and indexed in FAISS for cosine similarity search.
2. The top-k most similar company-years are retrieved for the target, ranked by embedding similarity, and enriched with shared risk themes and 1–2 years of lookahead financial data.

Comparison and Reporing: (In progress)

1. Loads all company-year vectors from Agent 2 into FAISS index for similarity search
2. Retrieves top unique peer companies for a queried company ticker
3. Apply lookahead retrieval & fetch the following years filing for matched companies
4. Generate grounded analysis that with risk signals, financial trends, and peer comparison with every claim mapped to source filings.

Contributions:
Data Curation — Sumit Kanu, Sharan Giri
Finding Peers — Om Mane
Comparison & Reporting — Sumit Kanu, Sharan Giri

5. Experimental Results

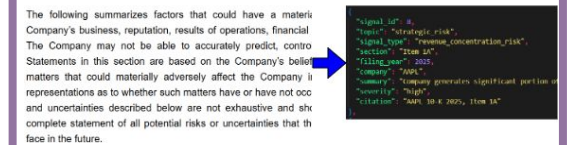


Fig. Extraction of risk signals with topic, severity and metadata.

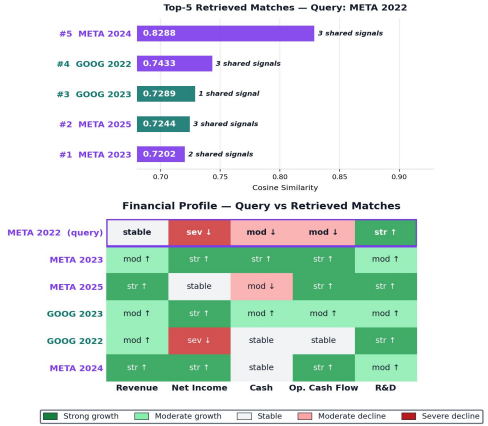


Fig. Top-5 FAISS-retrieved peers for META 2022 ranked by embedding similarity, with financial profile comparison across key metrics.

6. Future Work

Scale the reference corpus to ~15 verified companies using the full ingestion and extraction pipeline, enabling statistically meaningful cross-company risk trajectory comparison.